

A
Project Report on
**Predicting Digital Payment Platforms Adoption by Retail Vendors
Using Machine Learning Techniques**
Submitted in partial fulfilment of the requirements
for the award of the degree of
BACHELOR OF TECHNOLOGY
In
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

Ramineni Poojitha	22AK1A3357
Gurram Sidhardha	22AK1A3376
Mannuru Prudueeswar	22AK1A3363
Banka Varun	22AK1A3387

Under the guidance of

Mrs. S. Prathima, M.Tech.,
Assistant Professor



ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES
(AUTONOMOUS)

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ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



CERTIFICATE

Certified that this is a bonafide record of the Project Report entitled, **“Predicting Digital Payment Platforms Adoption by Retail Vendors Using Machine Learning Techniques”**, done by *Ramineni Poojitha (22AK1A3357)*, *Gurram Sidhardha (22AK1A3376)*, *Mannuru Prudueeswar (22AK1A3363)*, *Banka Varun (22AK1A3387)* is being submitted in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in **ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** to the Annamacharya Institute of technology and Sciences, Tirupati, during the academic year 2025-26.

Signature of the supervisor

Mrs. S. Prathima, M.Tech.,

Assistant professor,

Department of AI&ML,

AITS, Tirupati.

Signature of Head of the Department

Dr. C. Siva Balaji Yadav, M.Tech, Ph.D.,

Professor and HOD,

Department of AI&ML,

AITS, Tirupati.

DATE:

INTERNAL EXAMINER

EXTERNAL EXAMINER

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COMPUTER SCIENCE AND ENGINEERING



DECLARATION

We hereby declare that the project titled **“Predicting Digital Payment Platforms Adoption by Retail Vendors Using Machine Learning Techniques”** is a genuine project work carried out by us, in B.Tech (Artificial Intelligence and Machine Learning) course in Annamacharya Institute of Technology And Sciences and has not been submitted to any other course or university for the award of our degree by us.

Ramineni Poojitha	22AK1A3357
Gurram Sidhardha	22AK1A3376
Mannuru Prudueeswar	22AK1A3363
Banka Varun	22AK1A3387

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Ramineni Poojitha
Gurram Sidhardha
Mannuru Prudueeswar
Banka Varun

22AK1A3357
22AK1A3376
22AK1A3363
22AK1A3387

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S.NO	PUBLICATIONS
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3.	International Journal Certificates

ABSTRACT

The adoption of digital payment platforms by retail vendors has become a pivotal transformation in India's payment landscape. This project utilizes machine learning techniques to predict whether a retail vendor is likely to adopt digital payment systems such as Paytm, Google Pay, PhonePe, and Amazon Pay. Data was collected from 270 merchants across 10 major retail markets in Bangalore, targeting both organized and unorganized sectors. Through a comprehensive analytical approach involving descriptive statistics, exploratory factor analysis (EFA), and Support Vector Machine (SVM) modeling, the system identifies core factors influencing adoption: perceived usefulness, social influence, and compatibility. Additionally, the system evaluates customer satisfaction across platforms and forecasts future digital payment adoption trends using the ARIMA time series model. This integrated platform consists of three major modules: user (vendor interaction), admin (data management and analysis), and machine learning (prediction and forecasting). The system not only aids in understanding the behavior of retail vendors but also empowers policymakers, fintech providers, and entrepreneurs with actionable insights to improve the digital payments ecosystem.

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LIST OF ABBREVIATIONS

Short Form	Abbreviation
ML	Machine Learning
AI	Artificial Intelligence
SVM	Support Vector Machine
ARIMA	Auto Regressive Integrated Moving Average
EFA	Exploratory Factor Analysis
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
HTTP	Hypertext Transfer Protocol

1. INTRODUCTION

1.1 INTRODUCTION

1.1.1 Background and Motivation

A significant aspect of modern financial systems is the rapid adoption of digital payment technologies, which enable fast, secure, and convenient transactions without the need for physical cash. Digital payment platforms such as Paytm, Google Pay, PhonePe, and Amazon Pay have transformed the retail ecosystem by offering seamless payment solutions to both customers and vendors. These platforms enhance transaction efficiency, transparency, and accessibility, making them an essential component of today's digital economy.

The increasing penetration of smartphones and internet connectivity has further accelerated the shift toward cashless transactions. However, despite the availability and advantages of digital payment systems, many small and medium retail vendors are still hesitant to adopt these technologies. This hesitation is influenced by various factors such as perceived usefulness, ease of use, trust, social influence, and compatibility with existing business processes .

Understanding and predicting vendor adoption behaviour is crucial for improving the digital payment ecosystem. Traditional statistical approaches have limitations in capturing complex relationships among multiple influencing factors. With the advancement of machine learning, it is now possible to analyse large datasets, identify hidden patterns, and predict adoption behaviour more accurately. Techniques such as Support Vector Machine (SVM) and time-series models like ARIMA provide robust solutions for classification and forecasting tasks.

This project focuses on developing a machine learning-based system to predict the likelihood of digital payment adoption among retail vendors. By integrating statistical analysis with predictive modelling, the system aims to provide valuable insights to policymakers, fintech companies, and business stakeholders. Ultimately, this contributes to enhancing financial inclusion, improving vendor efficiency, and promoting the widespread adoption of digital payment technologies.

1.1.2 Problem Statement

The adoption of digital payment systems among retail vendors is influenced by multiple behavioral, technological, and environmental factors. Despite the benefits offered by digital platforms, many vendors face challenges such as lack of awareness, trust issues, technical barriers, and compatibility concerns with existing business practices. Traditional analytical methods are often insufficient to accurately predict adoption behavior due to their inability to handle complex and non-linear relationships among variables.

There is a need for a robust and efficient system that can analyze vendor data and predict the likelihood of digital payment adoption with higher accuracy. Machine learning techniques provide a promising solution by enabling automatic pattern recognition and predictive analysis. However, designing a system that ensures accuracy, scalability, and real-time usability remains a challenge.

This project aims to develop a machine learning-based prediction system using techniques such as Support Vector Machine (SVM) and ARIMA to classify vendors based on their adoption likelihood and forecast future trends. The goal is to build a reliable and scalable solution that supports decision-making and promotes the growth of digital payment ecosystems.

1.1.3 Objectives

1.3.1 Develop a Machine Learning-Based Prediction System for Digital Payment Adoption:

The primary objective of this project is to design and implement a predictive model using machine learning techniques, particularly Support Vector Machine (SVM), to classify retail vendors based on their likelihood of adopting digital payment platforms. The system aims to achieve high accuracy by analyzing key behavioural and operational factors.

1.3.2 Identify Key Factors Influencing Vendor Adoption:

The project aims to analyze and identify the major factors affecting digital payment adoption, such as perceived usefulness, social influence, and compatibility, using statistical techniques

like Exploratory Factor Analysis (EFA). This helps in understanding vendor behaviour and improving prediction accuracy.

1.3.3 Enable Data-Driven Decision Support for Stakeholders:

Another objective is to provide meaningful insights for policymakers, fintech companies, and business owners by predicting adoption trends and vendor behavior. This supports better decision-making and strategy development for increasing digital payment adoption.

1.3.4 Forecast Future Digital Payment Trends:

The system also aims to incorporate time-series forecasting using ARIMA models to predict future trends in digital payment adoption. This helps stakeholders anticipate market changes and plan accordingly.

1.3.5 Develop a User-Friendly and Scalable System:

The project focuses on building an integrated platform with user, admin, and machine learning modules, ensuring ease of use, scalability, and real-time prediction capabilities for practical implementation.

1.2 EXISTING SYSTEM

The existing approach to analyzing digital payment platform adoption among retail vendors is primarily based on traditional statistical and survey-based methods. In this system, data is collected from vendors through questionnaires and surveys to understand their behaviour toward digital payment systems. Researchers analyze this data using conventional techniques such as descriptive statistics, regression analysis, and theoretical models like the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT).

The process begins with gathering vendor responses related to factors such as perceived usefulness, ease of use, social influence, and compatibility. These factors are then analysed to identify patterns and relationships that influence adoption behaviour. The results are interpreted to explain why vendors adopt or do not adopt digital payment platforms. However, this system mainly focuses on understanding past and present behaviour rather than predicting future adoption trends.

One of the major limitations of the existing system is that it relies heavily on traditional statistical techniques. These methods are limited in their ability to capture complex and non-linear relationships among multiple influencing factors. As digital payment adoption involves various behavioral, technological, and social aspects, simple statistical models may not provide accurate insights.

Another significant issue is that the existing system is mostly explanatory in nature. It explains the factors affecting adoption but does not provide predictive capabilities. This makes it difficult for stakeholders to identify potential adopters or forecast future trends in digital payment usage.

Additionally, the system does not effectively integrate different analytical components. Customer satisfaction analysis and trend forecasting are often handled separately, which reduces the overall effectiveness of the system. There is no unified platform that combines all these aspects into a single framework.

Furthermore, the existing system is less suitable for handling large and multi-dimensional datasets. As the volume and complexity of data increase, traditional methods become inefficient and difficult to manage. This limits scalability and reduces the practical applicability of the system in real-world scenarios.

Overall, the existing system provides basic insights into digital payment adoption but lacks predictive power, scalability, and integration, highlighting the need for advanced machine learning-based solutions.

1.3 DISADVANTAGES OF EXISTING SYSTEM

- **Limited Predictive Capability**

The existing system mainly explains past data and cannot accurately predict future digital payment adoption.

- **Inability to Handle Complex Relationships**

Traditional methods fail to capture complex and non-linear relationships between multiple influencing factors.

- **Mostly Explanatory Approach**

The system focuses on understanding behavior rather than providing actionable predictions.

- **Lack of Machine Learning Integration**

Advanced machine learning techniques are not used, reducing analytical efficiency and accuracy.

- **Poor Trend Forecasting**

The system does not effectively forecast future adoption trends, limiting strategic planning.

- **Separate Customer Satisfaction Analysis**

Customer satisfaction is analysed separately, leading to fragmented insights.

- **Low Scalability with Large Data**

The system is not efficient when handling large and multi-dimensional datasets.

- **Inefficient Data Utilization**

Available data is not fully utilized to extract deeper insights and patterns.

1.4 PROPOSED SYSTEM

The proposed system introduces an advanced machine learning-based framework to predict digital payment platform adoption among retail vendors more effectively. Unlike the existing system, this approach integrates statistical analysis with machine learning techniques to provide both explanatory and predictive insights.

In this system, data is collected from retail vendors through structured questionnaires covering various factors such as perceived usefulness, social influence, and compatibility. The collected data is then pre-processed and analysed using Exploratory Factor Analysis (EFA) to identify the most significant factors influencing adoption .

These identified factors are used as input features for a Support Vector Machine (SVM) model, which classifies vendors based on their likelihood of adopting digital payment platforms. The SVM model is capable of handling complex and non-linear relationships, making it more accurate and reliable compared to traditional methods.

Additionally, the system incorporates ARIMA (Autoregressive Integrated Moving Average) time-series models to forecast future digital payment adoption trends. This allows stakeholders to gain insights into future market behaviour and plan strategies accordingly.

The proposed system also integrates customer satisfaction analysis across different digital payment platforms, providing a comprehensive understanding of vendor preferences and experiences. All these components are combined into a unified framework, ensuring better efficiency and scalability.

Overall, the proposed system provides a robust, integrated, and data-driven solution for analyzing and predicting digital payment adoption among retail vendors.

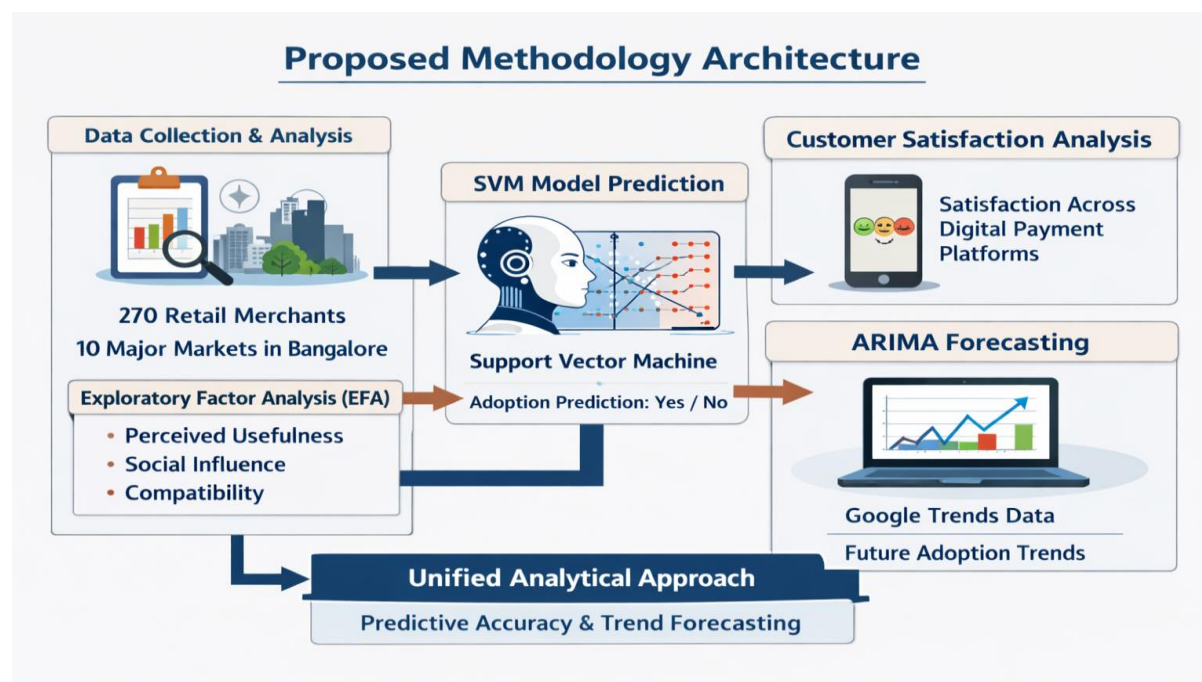


FIG.1.1 .PROPOSED SYSTEM ARCHITECTURE

1.5 ADVANTAGES OF PROPOSED SYSTEM

- **Integration of ML and Statistical Methods**

Combines machine learning with statistical analysis for better understanding and prediction.

- **Accurate Adoption Prediction**

Uses models like SVM to predict whether vendors will adopt digital payment systems.

- **Handles Complex Multi-Factor Data**

Effectively captures complex relationships between multiple influencing factors.

- **Identification of Key Influencing Factors**

Uses EFA to identify important factors such as usefulness, social influence, and compatibility.

- **Improved Model Performance (SVM)**

SVM provides better classification accuracy and reliable prediction results.

- **Customer Satisfaction Evaluation**

Analyses satisfaction across different digital payment platforms.

- **Future Trend Forecasting (ARIMA)**

Uses ARIMA model to predict future digital payment adoption trends.

- **Supports Data-Driven Decisions**

Provides insights that help stakeholders make informed decisions.

- **High Scalability and Efficiency**

The system can handle large datasets and works efficiently in real-world application

2. ANALYSIS

2.1 INTRODUCTION

With the rise of fintech innovation and increasing smartphone penetration, digital payment platforms have revolutionized the way retail transactions are conducted in India. Vendors are progressively shifting from traditional cash-based models to convenient, secure, and efficient digital alternatives. However, despite this trend, a significant portion of small and medium-sized vendors remains reluctant or inconsistent in adopting these platforms. The decision to adopt digital payment systems depends on multiple factors, including perceived ease of use, usefulness, social influence, compatibility, and trust. This study delves deep into those behavioural and technical factors to uncover what drives adoption and what hinders it. The city of Bangalore, known for its diverse retail market, provides the ideal environment for such a study. This project introduces a comprehensive system that combines statistical models with machine learning algorithms to predict digital payment adoption. Through structured modules and real-time analysis, the system helps retail vendors gain clarity on their digital readiness while offering strategic insights for improving fintech engagement across India's retail ecosystem.

2.2 SOFTWARE REQUIREMENTS SPECIFICATION

2.2.1 *Software Requirements*

Python IDE:

There are many IDEs available for Python development such as PyCharm, Visual Studio Code, Jupyter Notebook, Spyder, and Anaconda. Among these, Visual Studio Code and Jupyter Notebook are widely used due to their simplicity and strong support for data analysis. In our project, Visual Studio Code and Jupyter Notebook/Google Colab are used as they provide better support for machine learning model development and data visualization.

Python Packages:

NumPy, Pandas, Scikit-learn, Statsmodels, Matplotlib, Seaborn, Django, Pickle.

Python Libraries:

NumPy and Pandas are used for data preprocessing and manipulation, Scikit-learn is used for implementing machine learning models like SVM, Statsmodels is used for

ARIMA forecasting, Matplotlib and Seaborn are used for visualization, Django is used for web interface development, and Pickle is used for saving and loading trained models.

 Data Analysis Techniques:

Exploratory Data Analysis (EDA), Feature Selection, Exploratory Factor Analysis (EFA), Data Preprocessing, Normalization.

 Machine Learning Algorithms (Used):

Support Vector Machine (SVM), ARIMA (AutoRegressive Integrated Moving Average).

2.2.2: Hardware Components:

 **Processor:** Intel Core i3 or higher

 **Hard Disk:** Minimum 10 GB free storage

 **Memory:** At least 4 GB RAM

 **Input/Output Devices:** Keyboard, Mouse, Display Screen required for system interaction

2.3: FLOWCHART

The flowchart of the Digital Payment Adoption Prediction System represents the step-by-step workflow followed by the system to analyse vendor data and predict the likelihood of adopting digital payment platforms.

1. User Login / Authentication: The process begins when the user (admin or vendor) accesses the system through a web-based application. The user first logs into the system by providing valid credentials. Once authenticated, the system allows access to the data input interface where vendor-related information is entered.

2. Vendor Data Input & Storage: After successful login, the user proceeds to input vendor data such as demographic details, business type, and factors influencing digital payment adoption like perceived usefulness, social influence, and compatibility. This data is then

stored in the system database for further processing and analysis. The system ensures that all required fields are properly filled before proceeding to the next step.

3. Data Preprocessing: Once the data is collected, the system performs data preprocessing. In this stage, the data is cleaned by removing missing or inconsistent values, and necessary transformations such as normalization and encoding are applied. This step ensures that the dataset is in a suitable format for machine learning analysis.

4. Feature Selection & Analysis (EFA): After preprocessing, the system performs feature selection and analysis. Techniques such as Exploratory Factor Analysis (EFA) are used to identify the most important factors influencing digital payment adoption. These selected features are then used as input for the machine learning model.

5. Model Training (SVM): Next, the system moves to the model training phase where the Support Vector Machine (SVM) algorithm is applied. The model is trained using the processed dataset to learn patterns and relationships between input features and adoption behaviour. Once training is completed, the model is evaluated using performance metrics such as accuracy and precision to ensure reliability.

6. Adoption Prediction: After successful training and evaluation, the system proceeds to the prediction phase. When new vendor data is provided, the trained model analyses the input and predicts whether the vendor is likely to adopt digital payment systems. The prediction result is then displayed to the user through the interface.

7. Trend Forecasting (ARIMA): In addition to classification, the system also performs trend forecasting using the ARIMA model. This step analyses historical data to predict future trends in digital payment adoption. It helps stakeholders understand potential growth patterns and make informed decisions.

8. Result Visualization & Storage: Finally, the system presents the results in the form of visualizations such as graphs and charts. These visual outputs help users easily interpret the prediction results and trends. All results are stored in the system for future reference and analysis.

9. Conclusion of Workflow: Overall, the flowchart represents a structured and automated process that starts from data input and ends with prediction and visualization, ensuring accurate, efficient, and data-driven decision-making for digital payment adoption.

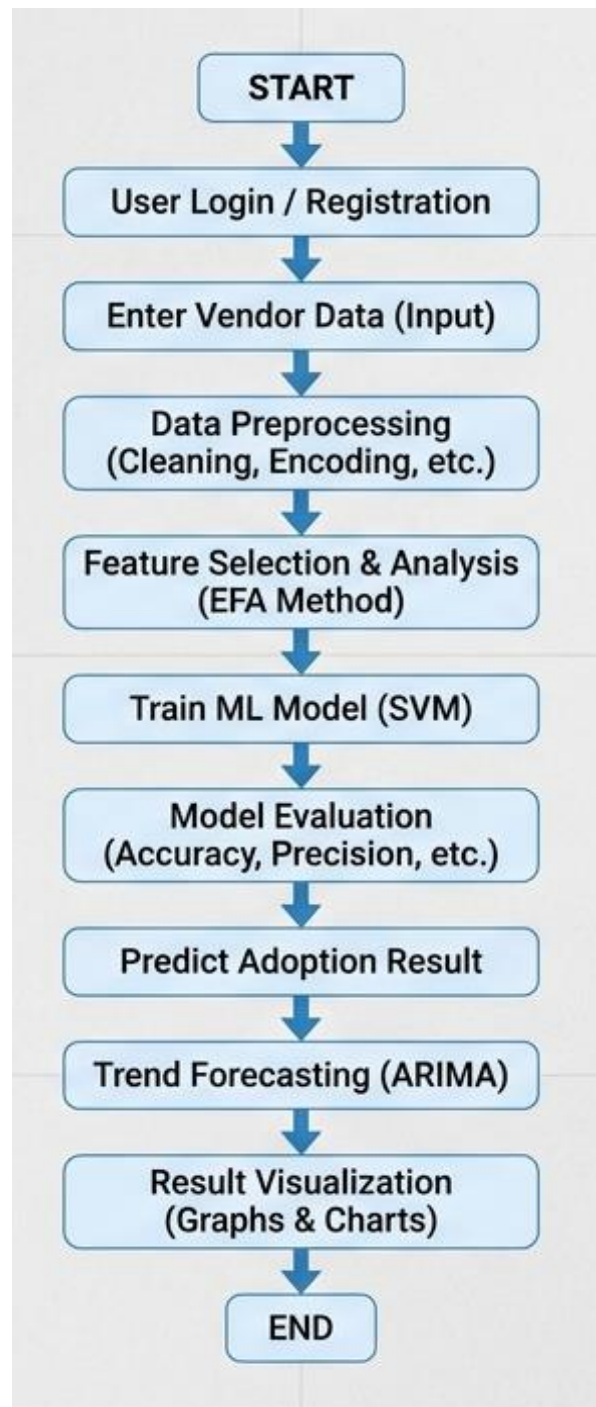


Fig 2.1: Vendor Adoption Process Flowchart

3. DESIGN

3.1 STEPS INVOLVED IN DIGITAL PAYMENT ADOPTION PREDICTION

3.1.1 Data Gathering and Data Pre-processing

- Collect dataset from retail vendors using structured questionnaires (demographic, business, and behavioural data).
- Include factors such as perceived usefulness, social influence, trust, and compatibility.
- Remove incomplete and inconsistent data entries.
- Handle missing values and clean the dataset.
- Encode categorical data into numerical format.
- Normalize numerical features to ensure uniform scaling.
- Perform feature selection using statistical techniques like Exploratory Factor Analysis (EFA).
- Split dataset into training and validation sets (80% training, 20% validation).

3.1.2 Training the Model Using Machine Learning Algorithms

- Train the model using Support Vector Machine (SVM) for classification of vendors.
- Identify whether a vendor is likely to adopt digital payment or not.
- Use kernel functions to handle non-linear relationships.
- Tune hyperparameters such as kernel type and regularization parameter.
- Evaluate model performance using accuracy, precision, recall, and F1-score.
- Integrate ARIMA model for forecasting future adoption trends.
- Analyse time-series data to predict future growth in digital payments.
- Validate model using performance metrics like MAE, MSE, and R^2 score.
- Improve model performance by refining features and optimizing parameters.

3.1.3 Final Prediction Model Integration

The final trained model (SVM + ARIMA) is integrated into a web-based system using Django framework. It takes vendor input data, processes it, and predicts whether the vendor will adopt digital payments. The system displays prediction results such as “Likely to Adopt” or “Not Likely to Adopt” along with analytical insights.

3.2 SYSTEM MODULES

Data Collection Module:

- Collect vendor data using structured surveys.
- Include demographic, business, and behavioural attributes.
- Perform preprocessing like cleaning, encoding, and normalization.

Feature Extraction:

- Use Exploratory Factor Analysis (EFA) to identify important features.
- Extract key factors such as usefulness, social influence, and compatibility.
- Reduce dimensionality and remove redundant features.

Model Training & Validation:

- Train SVM model using processed dataset.
- Split dataset into training and validation sets.
- Tune hyperparameters for better performance.
- Validate using metrics like accuracy, precision, recall, and F1-score.

Prediction & Classification:

- Classify vendors into adopters and non-adopters.
- Use trained SVM model for prediction.
- Generate real-time predictions based on user input.

System Integration & Web Interface:

- Develop web application using Django framework.
- Provide user login and data input interface.
- Store user data and predictions in database.
- Display prediction results through frontend interface.

Forecasting & Analytics:

- Use ARIMA model for time-series forecasting.
- Predict future digital payment adoption trends.
- Provide insights for decision-making and policy planning.

Performance Evaluation & Optimization:

- Evaluate model using MAE, MSE, and R^2 score.
- Analyze prediction accuracy and errors.
- Improve system using better features and advanced algorithms.

- Compare models like SVM, Random Forest, and Gradient Boosting for better performance.

3.3 DFD/UML DIAGRAMS

3.3.1 DFD Diagrams

A DFD is a graphical representation of how the data flows through a system. Developing a DFD is one of the first steps carried out when developing an information system. DFD displays details like the data that is coming in and going out of the system, how the data travels through the system, and how the data will be stored in the system. However, the DFD does not contain information about the timing of the processes. The main components included in a DFD are processes, data stores, data flow, and external entities. When developing DFD diagrams, the context level DFD is drawn first. It displays how the entire system interacts with external data sources and data sinks. Next, a Level 0 DFD is developed by expanding the context level DFD. Level 0 DFD contains details of the sub-systems within the system and how the data flows through them. It also contains details about the data stores required within the system. Yourdon & Coad and Gane & Sarson are two notations that are used to draw DFD.

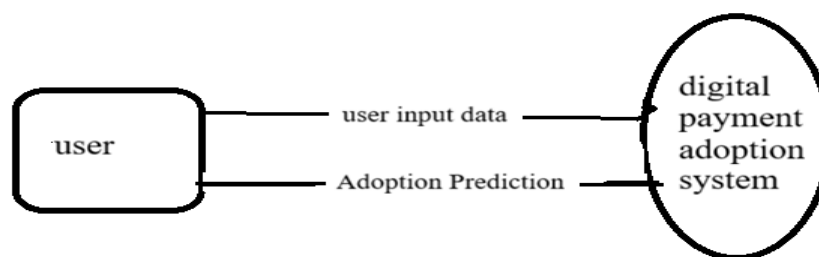


Fig 3.1: Level-0 DFD

The context diagram (DFD Level 0) for the Digital Payment Adoption Prediction System provides a high-level overview of how the system interacts with users and processes vendor-related data.

In this system, the primary external entities are the Retail Vendor and the Admin. The retail vendor provides input data such as demographic details, business characteristics, and

behavioural factors (like perceived usefulness, social influence, and compatibility) to the central system, the Digital Payment Adoption Prediction System.

The system processes this input using machine learning models such as Support Vector Machine (SVM) and forecasting techniques like ARIMA to analyse vendor behaviour and predict the likelihood of adopting digital payment platforms. The system then generates outputs in the form of adoption predictions, analytical insights, and future trend forecasts.

These results are sent back to the vendor in an understandable format, while the admin receives reports, analytics, and system insights for monitoring and decision-making.

The diagram abstracts internal processing details and focuses on the interaction between external entities and the system. It highlights how the system acts as an intelligent bridge between vendor data and predictive insights, enabling better decision-making and promoting digital payment adoption.

Future enhancements may include real-time data integration, advanced machine learning models, mobile-based interfaces, and integration with fintech platforms to improve prediction accuracy and usability.

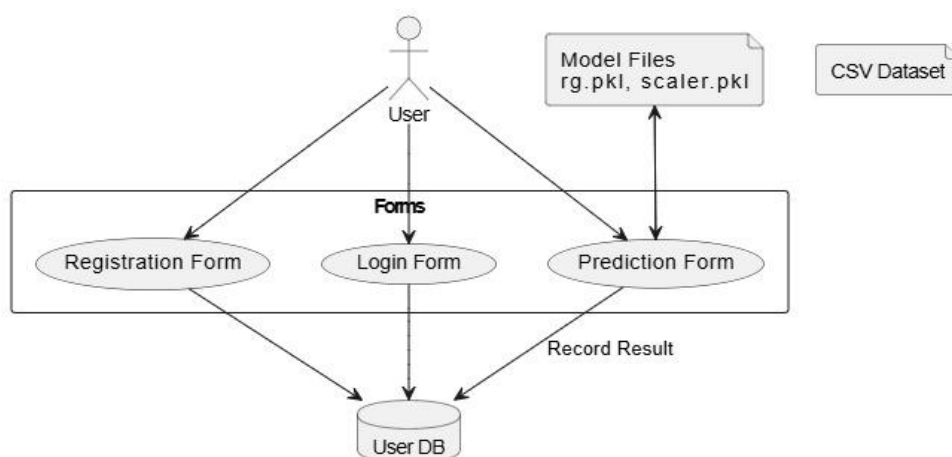


Fig 3.2: Level-1 DFD

The Level-1 Data Flow Diagram (DFD) for the Digital Payment Adoption Prediction System provides a detailed representation of the internal data flow and component interactions, expanding upon the high-level process shown in the Level-0 DFD.

In this system, the retail vendor provides input data such as demographic details, business information, and behavioural factors through a web-based interface. This input is captured and passed to the Data Collection and Preprocessing module, where the data is cleaned, missing values are handled, and categorical variables are encoded. The processed data is then normalized to ensure consistency and improve model performance.

The pre-processed data is forwarded to the Feature Extraction module, where important factors such as perceived usefulness, social influence, trust, and compatibility are identified using statistical techniques like Exploratory Factor Analysis (EFA). These refined features are then sent to the Machine Learning Model module, which acts as the core processing unit of the system.

Within this module, the Support Vector Machine (SVM) model analyzes the input features and classifies vendors into categories such as likely adopters or non-adopters of digital payment systems. Simultaneously, historical and trend-related data is passed to the ARIMA forecasting module, which predicts future digital payment adoption trends based on time-series analysis.

The outputs from both the prediction and forecasting modules are then sent to the Results and Reporting module, where the system generates meaningful insights, adoption predictions, and analytical reports. These results are displayed to the vendor through the user interface and are also made available to the admin for monitoring and decision-making.

Additionally, the system stores processed data and prediction results in a database for future reference, analysis, and model improvement. The system may also update models periodically using new data to enhance accuracy and performance.

This Level-1 diagram illustrates the complete workflow of the system, including data collection, preprocessing, feature extraction, machine learning-based classification, forecasting, and result generation. It highlights how the integration of statistical analysis, machine learning, and web technologies enables intelligent prediction of digital payment adoption, supporting better decision-making and promoting digital transformation in the retail sector.

3.3.2 UML Diagrams

UML is a modelling language used in object-oriented software design. UML provides capabilities to specify and visualize the components that make up a software system. UML diagrams mainly represent the structural view and the behavioral view of a system. The structural view of the system is represented using diagrams like class diagrams, composite structure diagrams, etc. A dynamic view of the system is represented using diagrams such as sequence diagrams, activity diagrams, etc. UML version 2.2 includes fourteen diagrams, which include seven diagrams for representing the structural view and the other seven representing the behavioral view. Among the seven behavioral diagrams, four diagrams can be used to represent interactions with the system. There are tools that can be used for UML modelling, such as IBM Rational Rose.

3.3.2.1 Class Diagram

Class diagram is a static diagram. It represents the static view of an application. Class diagram is not only used for visualizing, describing, and documenting different aspects of a system but also for constructing executable code of the software application. Class diagram describes the attributes and operations of a class and also the constraints imposed on the system. The class diagrams are widely used in the modelling of object-oriented systems because they are the only UML diagrams, which can be mapped directly with object-oriented languages. Class diagram shows a collection of classes, interfaces, associations, collaborations, and constraints. It is also known as a structural diagram. The purpose of class diagram is to model the static view of an application. Class diagrams are the only diagrams which can be directly mapped with object-oriented languages and thus widely used at the time of construction. UML diagrams like activity diagram, sequence diagram can only give the sequence flow of the application, however class diagram is a bit different. It is the most popular UML diagram in the coder community. The purpose of the class diagram can be summarized as

- Analysis and design of the static view of an application.
- Describe the responsibilities of a system.
- Base for component and deployment diagrams.
- Forward and reverse engineering. The class diagram helps construct the code for the

software application development.

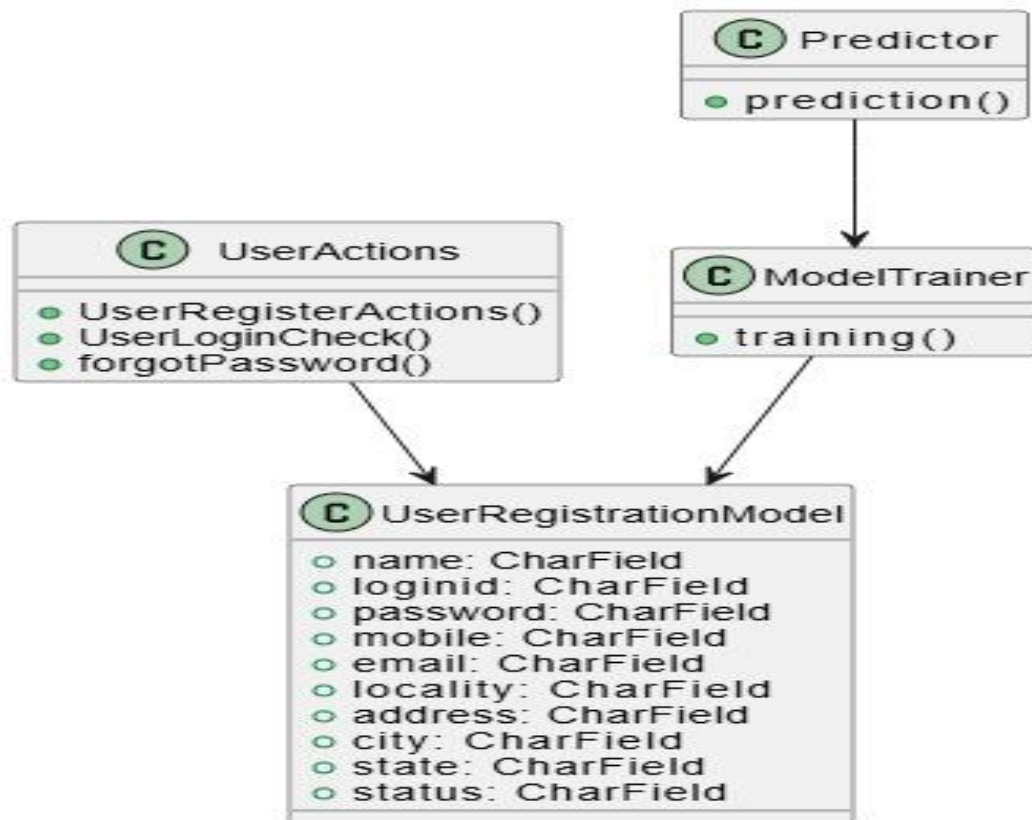


Fig 3.3: Class Diagram

3.3.2.2 Use Case Diagram

In UML, use-case diagrams model the behaviour of a system and help to capture the requirements of the system. Use-case diagrams describe the high-level functions and scope of a system. These diagrams also identify the interactions between the system and its actors. The use cases and actors in use-case diagrams describe what the system does and how the actors use it, but not how the system operates internally. Use-case diagrams illustrate and define the context and requirements of either an entire system or the important parts of the system. You can model a complex system with a single use-case diagram or create many use-case diagrams to model the components of the system. You would typically develop use-case diagrams in the early phases of a project and refer to them throughout the development process. Use cases are represented with a labeled oval shape. Stick figures represent actors in the process, and the actor's participation in the system is modelled with a line between the

actor and the use case. To depict the system boundary, draw a box around the use case itself.

UML Use case diagrams are ideal for:

- ✚ Representing the goals of system-user interactions.
- ✚ Defining and organizing functional requirements in a system.
- ✚ Specifying the context and requirements of a system.
- ✚ Modelling basic flow of events in a use case.

An effective use case diagram can help your team discuss and represent:

- ✚ Scenarios in which your system or application interacts with people, organizations, or external systems.
- ✚ Goals that your system or application helps those entities (known as actors) achieve the scope of your system.

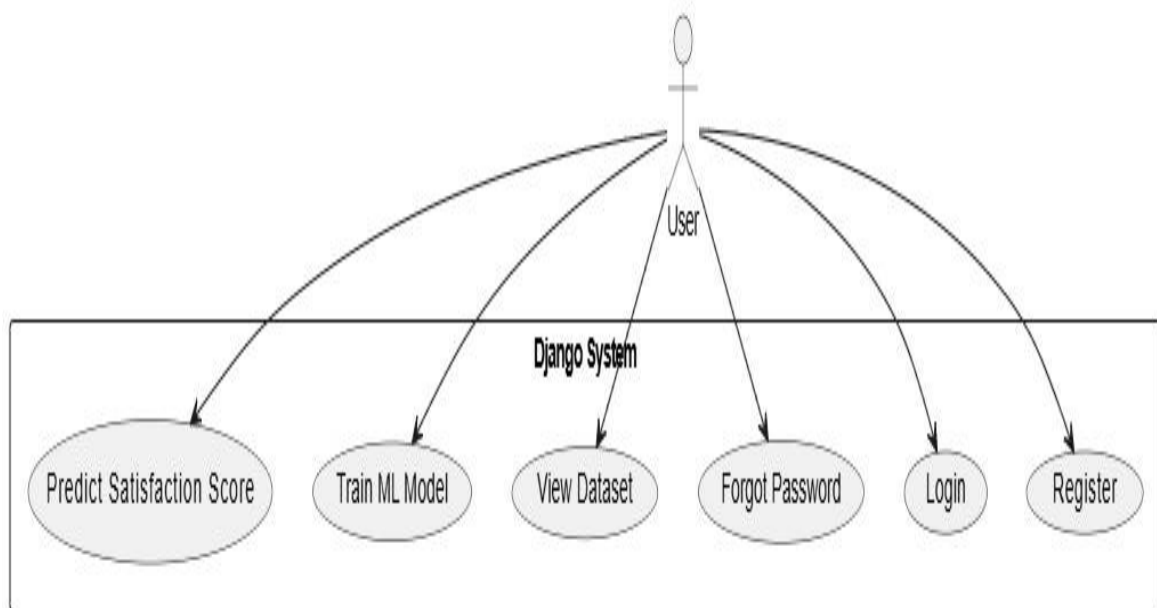


Fig 3.4: Use Case Diagram

3.3.2.3 Sequence Diagram

A sequence diagram is a type of interaction diagram because it describes how and in what order a group of objects works together. These diagrams are used by software developers and business professionals to understand requirements for a new system or to document an existing process. Sequence diagrams are sometimes known as event diagrams or event scenarios. Sequence diagrams can be useful references for businesses and other organizations. Try drawing a sequence diagram to:

- 🚦 Represent the details of a UML use case.
- 🚦 Model the logic of a sophisticated procedure, function, or operation.
- 🚦 See how objects and components interact with each other to complete a process.
- 🚦 Plan and understand the detailed functionality of an existing or future scenario.

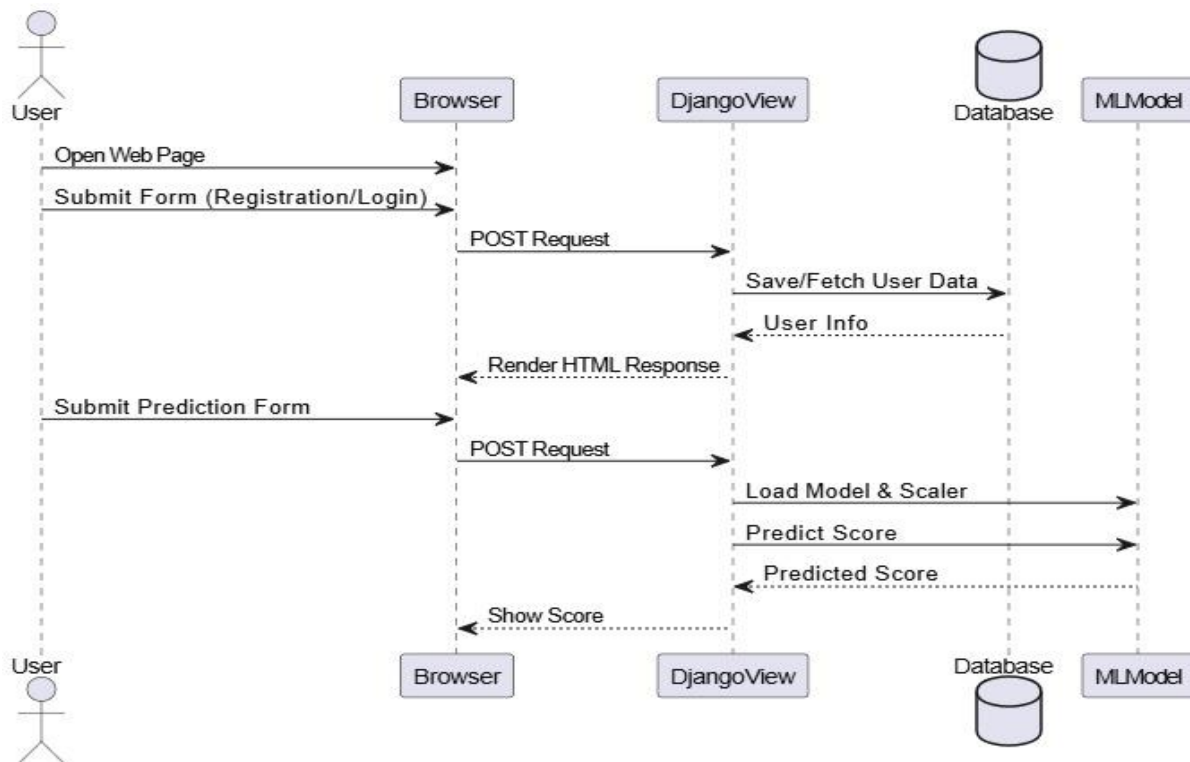


Fig 3.5: Sequence diagram

3.3.2.4 Deployment Diagram

A deployment diagram visualizes the physical deployment of software components in a system. It illustrates how software and hardware components are distributed across different nodes in a network and how they interact with each other to form a complete system. The primary purpose of a deployment diagram is to show the mapping of software components to the hardware components on which they run.

- 🚦 Nodes represent physical entities, such as hardware devices or servers, where components are deployed. They could be a computer, a server, a router, or any other device that can host software.
- 🚦 Components represent the software elements or modules that are deployed on nodes. These can be executable files, libraries, or other software artifacts.

- ✚ Artifacts are physical files or data that are used or produced by components. They can include databases, configuration files, or any other data storage.
- ✚ Associations show the relationships and connections between nodes and components. They indicate how components are deployed on specific nodes and how nodes interact with each other.
- ✚ Communication paths depict the channels or network connections through which nodes communicate with each other. They show the flow of messages or data between

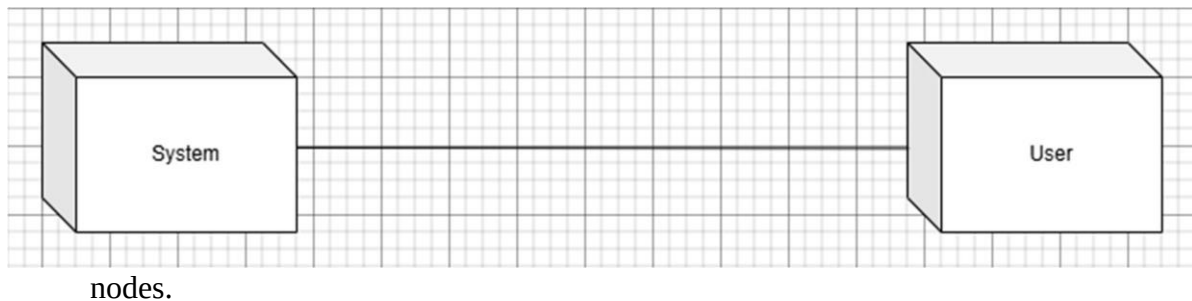


Fig 3.6: Deployment diagram

3.3.2.5 Activity Diagram

An activity diagram is a behavioral diagram, i.e. it depicts the behavior of a system. An activity diagram is basically a flowchart to represent the flow from one activity to another. The activity can be described as an operation of the system. The control flow is drawn from one operation to another. This flow can be sequential, branched, or concurrent. Activity diagrams deal with all types of flow control by using different elements such as fork, join, etc. The basic purposes of activity diagrams is similar to other four diagrams. It captures the dynamic behavior of the system. The other four diagrams are used to show the message flow from one object to another but the activity diagram is used to show message flow from one activity to another. Activity is a particular operation of the system. Activity diagrams are not only used for visualizing the dynamic nature of a system, but they are also used to construct the executable system by using forward and reverse engineering techniques. The only missing thing in the activity diagram is the message part. It does not show any message flow from one activity to another. Activity diagram is sometimes considered as the flowchart.

Although the diagrams look like a flowchart, they are not. It shows different flows such as parallel, branched, concurrent, and single. An activity diagram portrays the control flow from a start point to a finish point showing the various decision paths that exist while the activity is being executed. The purpose of an activity diagram can be described as –

- ✚ Draw the activity flow of a system.
- ✚ Describe the sequence from one activity to another.
- ✚ Describe the parallel, branched and concurrent flow of the system.

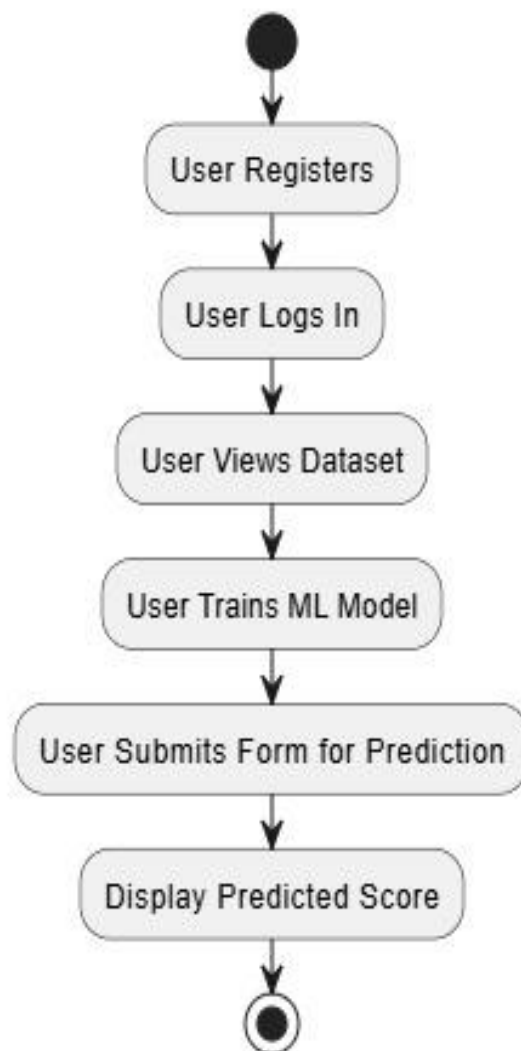


Fig 3.7: Activity Diagram

4. IMPLEMENTATION & DETAILS

4.1 INTRODUCTION

Digital payment adoption has become a crucial aspect of modern financial ecosystems, especially with the rapid growth of fintech technologies and mobile-based transactions. Traditional cash-based systems are gradually being replaced by digital payment platforms such as UPI, mobile wallets, and online banking, offering convenience, speed, and transparency. However, despite these advantages, many retail vendors still hesitate to adopt digital payment systems due to various behavioural, technological, and operational factors.

A Digital Payment Adoption Prediction System aims to analyse vendor behaviour and predict whether a retail vendor is likely to adopt digital payment methods. This system uses machine learning techniques to transform raw vendor data into meaningful insights, enabling better decision-making for businesses, policymakers, and fintech companies.

To achieve this, machine learning algorithms such as Support Vector Machine (SVM) are used for classification, while ARIMA models are used for forecasting future trends. These models can automatically learn patterns from vendor data without requiring manual rule-based analysis, making them highly effective for prediction tasks.

In this project, we propose a machine learning-based system that predicts digital payment adoption using vendor-related data such as demographic details, business characteristics, and behavioural factors like perceived usefulness, social influence, and compatibility. The system processes this data, trains predictive models, and generates insights regarding adoption likelihood.

Unlike traditional approaches that rely on statistical assumptions, this system adapts to real-world data patterns, improving accuracy and scalability. The primary objective is to provide a reliable, data-driven solution that supports digital transformation and enhances financial inclusion.

4.1.1 Process of Digital Payment Adoption Prediction

The Digital Payment Adoption Prediction System follows a systematic process to collect vendor data, analyse behavioural patterns, and predict the likelihood of adopting digital payment systems. The workflow of the system consists of several sequential steps, which are explained below in detail.

1. Vendor Registration

The first step in the system is vendor registration. In this stage, the retail vendor creates an account by entering details such as name, business type, location, contact information, and login credentials. The system validates the entered data and securely stores it in the database. This ensures that each vendor has a unique identity in the system and can access the platform.

2. Vendor Login

After registration, the vendor logs into the system using their registered credentials such as email or mobile number and password. The system verifies the login details and provides access to the vendor dashboard. This authentication mechanism ensures secure access to the system.

3. Data Input by Vendor

Once logged in, the vendor provides required input data related to their business and behaviour. This includes:

- Demographic details
- Business characteristics
- Technology usage
- Behavioural factors (usefulness, trust, social influence, compatibility)

This data acts as the primary input for prediction.

4. Data Preprocessing

After data submission, the system processes the input data to make it suitable for machine learning models. This includes:

- Cleaning inconsistent or missing data
- Encoding categorical variables into numerical values
- Normalizing numerical features
- Selecting important features using statistical methods

This step ensures data quality and improves prediction accuracy.

5. Feature Extraction and Selection

The system identifies the most relevant features influencing digital payment adoption.

Techniques like Exploratory Factor Analysis (EFA) are used to extract key factors such as:

- Perceived usefulness
- Social influence
- Trust
- Compatibility

This reduces unnecessary data and enhances model efficiency.

6. Model Processing (Prediction Engine)

The processed data is passed to the machine learning model, which acts as the core component of the system.

- The Support Vector Machine (SVM) model analyses the input data
- It classifies vendors into categories such as:
 - **Likely to Adopt**
 - **Not Likely to Adopt**
- Kernel functions are used to handle complex patterns in data

7. Forecasting Future Trends

In addition to classification, the system performs trend analysis using forecasting models.

- The ARIMA model analyses historical data
- Predicts future digital payment adoption trends
- Provides insights into growth patterns

8. Data Storage

All input data, processed data, and prediction results are stored in the database.

- Maintains records for future analysis
- Helps in improving model performance

- Enables admin to track vendor behaviour

9. Admin Monitoring and Analysis

The administrator can access system reports and analytics.

- View prediction results of multiple vendors
- Analyse adoption trends
- Make data-driven decisions
- Modify system parameters if required

10. Performance Evaluation

The system evaluates model performance using metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- MAE, MSE, and R^2 (for forecasting)

This ensures the reliability of predictions.

11. System Improvement and Updating

To maintain system efficiency:

- New data is continuously collected
- Model is retrained periodically
- Features are updated
- Performance is optimized

This ensures the system remains accurate and adaptable to changing trends.

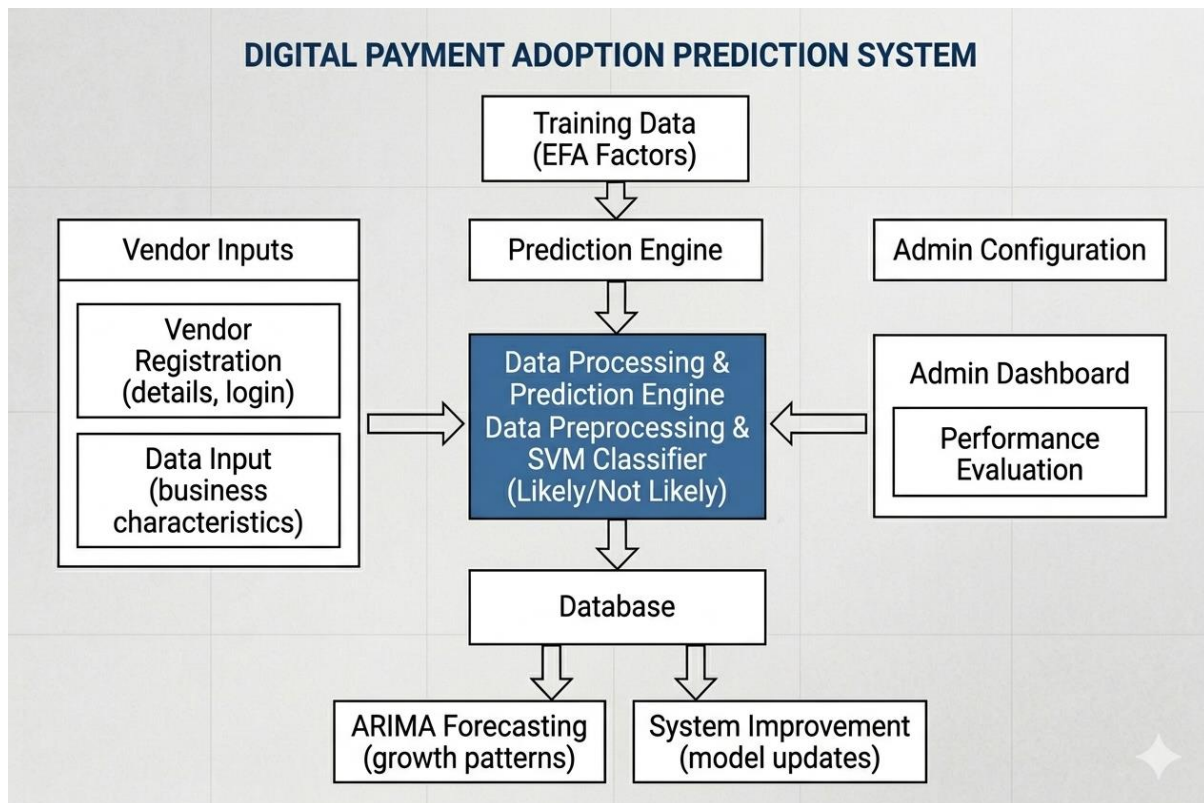


Fig 4.1: Representation of *Digital payment Aadoption prediction*

Sample Code:

```
import pandas as pd
import pickle
import os

from django.shortcuts import render
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# ----- TRAINING FUNCTION ----- #
def training(request):
```

Correct file path

```
file_path = os.path.join('media', 'DigitalPay_balanced_dataset (1).csv')  
df = pd.read_csv(file_path)
```

----- CLEANING -----

```
if 'Customer_ID' in df.columns:  
    df.drop(['Customer_ID'], axis=1, inplace=True)
```

```
y = df['Customer_Satisfaction_Score']
```

#USE ONLY 8 IMPORTANT FEATURES

```
X = df[[  
    "Age",  
    "Total_Transactions",  
    "Avg_Transaction_Value",  
    "Total_Spent",  
    "Active_Days",  
    "Last_Transaction_Days_Ago",  
    "Cashback_Received",  
    "Loyalty_Points_Earned"  
]]
```

----- SPLIT -----

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y,  
    test_size=0.2,  
    random_state=42,  
    stratify=y  
)
```

----- SCALING -----

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# ----- SVM(SVC MODEL) for classificaton ----- #
model = SVC(
    kernel='rbf',
    C=50,
    gamma=0.01,
    probability=True
)

model.fit(X_train_scaled, y_train)

# ----- SAVE FILES ----- #
pickle.dump(model, open('model.pkl', 'wb'))
pickle.dump(scaler, open('scaler.pkl', 'wb'))

# ----- Evaluation METRICS ----- #
y_pred = model.predict(X_test_scaled)

accuracy = round(accuracy_score(y_test, y_pred) * 100, 2)
precision = round(precision_score(y_test, y_pred) * 100, 2)
recall = round(recall_score(y_test, y_pred) * 100, 2)
f1 = round(f1_score(y_test, y_pred) * 100, 2)

return render(request, "users/training.html", {
    "accuracy": accuracy,
    "precision": precision,
    "recall": recall,
    "f1": f1
})
```

```
}}
```

```
# ----- PREDICTION FUNCTION ----- #
```

```
def prediction(request):
```

```
    result = None
```

```
    confidence = None
```

```
    if request.method == "POST":
```

```
        if not os.path.exists('model.pkl'):
```

```
            return render(request, 'users/prediction.html', {
```

```
                "result": " ⚠ Train model first",
```

```
                "confidence": 0
```

```
            })
```

```
    model = pickle.load(open('model.pkl', 'rb'))
```

```
    scaler = pickle.load(open('scaler.pkl', 'rb'))
```

```
    input_data = pd.DataFrame([[
```

```
        float(request.POST['Age']),
```

```
        float(request.POST['Total_Transactions']),
```

```
        float(request.POST['Avg_Transaction_Value']),
```

```
        float(request.POST['Total_Spent']),
```

```
        float(request.POST['Active_Days']),
```

```
        float(request.POST['Last_Transaction_Days_Ago']),
```

```
        float(request.POST['Cashback_Received']),
```

```
        float(request.POST['Loyalty_Points_Earned'])
```

```
    ]], columns=[
```

```
        "Age",
```

```
        "Total_Transactions",
```

```
        "Avg_Transaction_Value",
```



```
"Total_Spent",
"Active_Days",
"Last_Transaction_Days_Ago",
"Cashback_Received",
"Loyalty_Points_Earned"
])

input_scaled = scaler.transform(input_data)

# ----- PREDICT ----- #
pred = model.predict(input_scaled)[0]
prob = model.predict_proba(input_scaled)[0]

confidence = round(max(prob) * 100, 2)

if pred == 1:
    result = "Likely to Adopt"
else:
    result = "Unlikely to Adopt"

print("Prediction Probabilities:", prob)

return render(request, 'users/prediction.html', {
    "result": result,
    "confidence": confidence
})
```

4.2 EXPLANATION OF KEY FUNCTIONS

Digital Payment Adoption Prediction System is a powerful data-driven solution that enhances decision-making, financial inclusion, and business intelligence in the fintech domain. Below are the key features and applications where this system proves especially useful:

- **Data-Driven Decision Making**

The system uses machine learning techniques to analyse vendor data and generate accurate predictions. This helps businesses, policymakers, and fintech companies make informed decisions based on real insights rather than assumptions.

- **Prediction of Vendor Adoption Behaviour**

The system predicts whether a retail vendor is likely to adopt digital payment methods. By analysing behavioural and operational factors, it classifies vendors into adopters and non-adopters, helping stakeholders target the right audience.

- **Identification of Key Influencing Factors**

Using statistical techniques like Exploratory Factor Analysis (EFA), the system identifies critical factors such as perceived usefulness, social influence, trust, and compatibility. This helps in understanding what drives or restricts digital payment adoption.

- **Real-Time Prediction System**

The system provides real-time predictions through a web-based interface. Vendors can input their data and instantly receive results, making the system interactive and user-friendly.

- **Integration with Web Technologies**

The system is implemented using a web framework, allowing easy access through browsers. It includes features like user login, data entry forms, and result dashboards, ensuring smooth interaction between users and the system.

- **Forecasting Future Adoption Trends**

In addition to prediction, the system uses time-series models like ARIMA to forecast future digital payment adoption trends. This helps organizations understand growth patterns and plan strategies accordingly.

- **Support for Financial Inclusion**

The system helps identify vendors who are less likely to adopt digital payments and understand their challenges. This enables targeted awareness programs and policy-making to promote digital financial inclusion.

- **Scalable and Flexible Architecture**

The system is designed to handle large datasets and can be easily extended with advanced machine learning models. It can adapt to different regions, business types, and changing market conditions.

• **Performance Evaluation and Optimization**

The system evaluates model performance using metrics like accuracy, precision, recall, F1-score, MAE, MSE, and R^2 score. Continuous optimization ensures improved prediction accuracy over time.

• **Business and Policy Insights**

The system provides valuable insights for:

- Fintech companies to improve services
- Governments to design digital initiatives
- Businesses to enhance customer experience

• **Reduction of Manual Analysis**

Traditional analysis methods require manual effort and may not capture complex patterns. This system automates the entire process, saving time and increasing efficiency.

• **Future Enhancement Capabilities**

The system can be further enhanced by:

- Integrating advanced AI models (Random Forest, Neural Networks)
- Adding mobile-based applications
- Incorporating real-time data streams
- Expanding to other financial domains

4.3 METHOD OF IMPLEMENTATION

Machine Learning algorithms are utilized to implement the Digital Payment Adoption Prediction System due to their ability to analyse structured data and identify complex patterns in vendor behaviour. These algorithms are highly effective for classification and forecasting tasks, making them suitable for predicting whether a retail vendor is likely to adopt digital payment systems. The process begins with the collection of a comprehensive dataset containing vendor-related information such as demographic details, business characteristics, and behavioural factors. Each data record is labelled based on whether the vendor has adopted digital payment systems or not. This labelled dataset is then used to train the machine

learning model. The system primarily uses Support Vector Machine (SVM) for classification and ARIMA for time-series forecasting.

The SVM model consists of the following key components:

- **Input Features:** Vendor data such as usefulness, trust, social influence, and compatibility.
- **Kernel Functions:** Used to transform data into higher dimensions to handle non-linear relationships (e.g., linear, polynomial, RBF).
- **Hyperplane:** A decision boundary that separates vendors into different classes (adopters and non-adopters).
- **Support Vectors:** Critical data points that help define the optimal boundary.

The model is trained on the dataset where it learns to classify vendors by minimizing classification errors using optimization techniques.

Once the classification model is trained, a forecasting model is also applied:

- **ARIMA Model:** Used to analyse time-series data and predict future digital payment adoption trends.
- **Trend Analysis:** Helps in understanding growth patterns and future adoption rates.

After training and validation, the system can accurately predict vendor adoption behaviour and provide future insights, enabling better decision-making for businesses and policymakers.

4.3.1 Machine Learning Model

In this implementation, Support Vector Machine (SVM) is used for classification and ARIMA is used for forecasting digital payment adoption trends. These models are chosen due to their effectiveness in handling structured data and prediction tasks.

SVM Model Advantages:

- It can handle both linear and non-linear data using kernel functions.
- It is effective for small to medium-sized datasets.
- It provides high accuracy in classification problems.
- It works well in high-dimensional feature spaces.

ARIMA Model Advantages:

- It is suitable for time-series forecasting.
- It captures trends and patterns in historical data.

- It provides reliable predictions for future adoption trends.

Model Training Process:

The machine learning model is trained using vendor datasets, where the data undergoes preprocessing before being fed into the model.

- Data Preprocessing: Cleaning, encoding, and normalization of vendor data.
- Feature Selection: Identifying important features using statistical methods.
- Model Training: Training SVM model using labelled data.
- Validation: Evaluating model performance using test data.

Real-Time Prediction:

After training, the model is integrated into the system for real-time usage.

- Vendor inputs data through the web interface.
- The system preprocesses the input data.
- The SVM model predicts whether the vendor will adopt digital payments.
- The result is displayed instantly to the user.

4.3.2 Output Screens:

Screen 4.1: User Registration Page

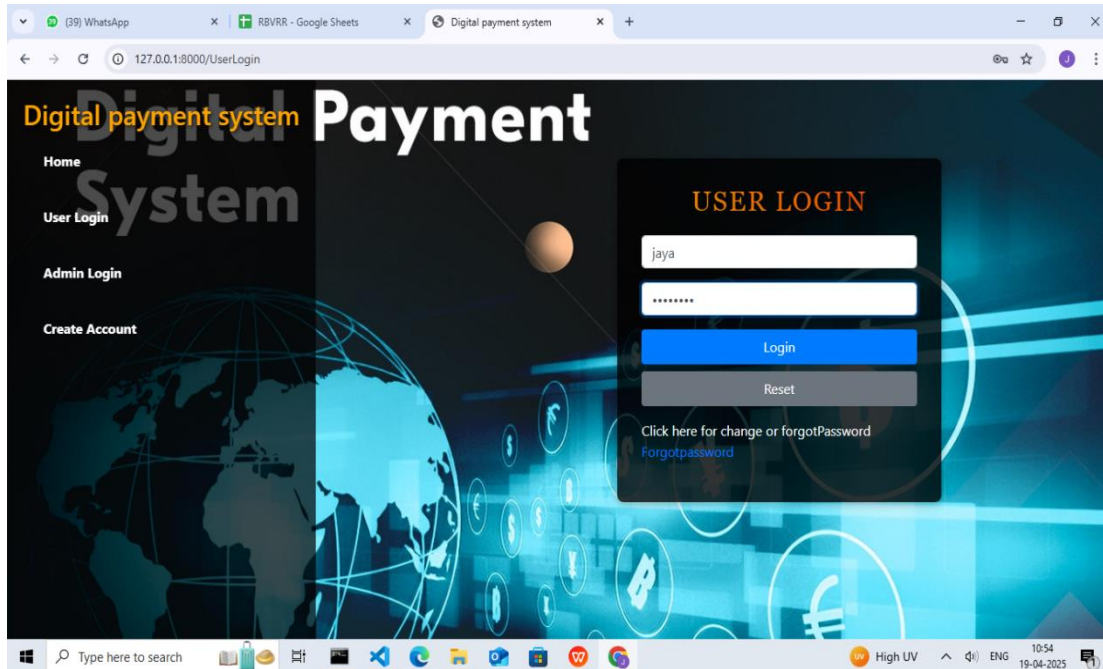
This is the page where the new user can register.

The screenshot shows a web browser window with the URL `127.0.0.1:8000/UserRegisterForm`. The page has a dark blue background with a globe and a large dollar sign. On the left, there is a sidebar with links: Home, User Login, Admin Login, and Create Account. The main content area is titled "REGISTER HERE" and contains a registration form with the following fields: Name (jaya), Login ID (jaya), Password (masked with asterisks), Mobile (7896543567), Email (j@gmail.com), Locality (hyd), and Address (empty). The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system tray on the right shows "High UV", volume, language (ENG), and the date/time (10:54, 19-04-2025).

Predicting Digital Payment Platforms Adoption by Retail Vendors Using Machine Learning Techniques

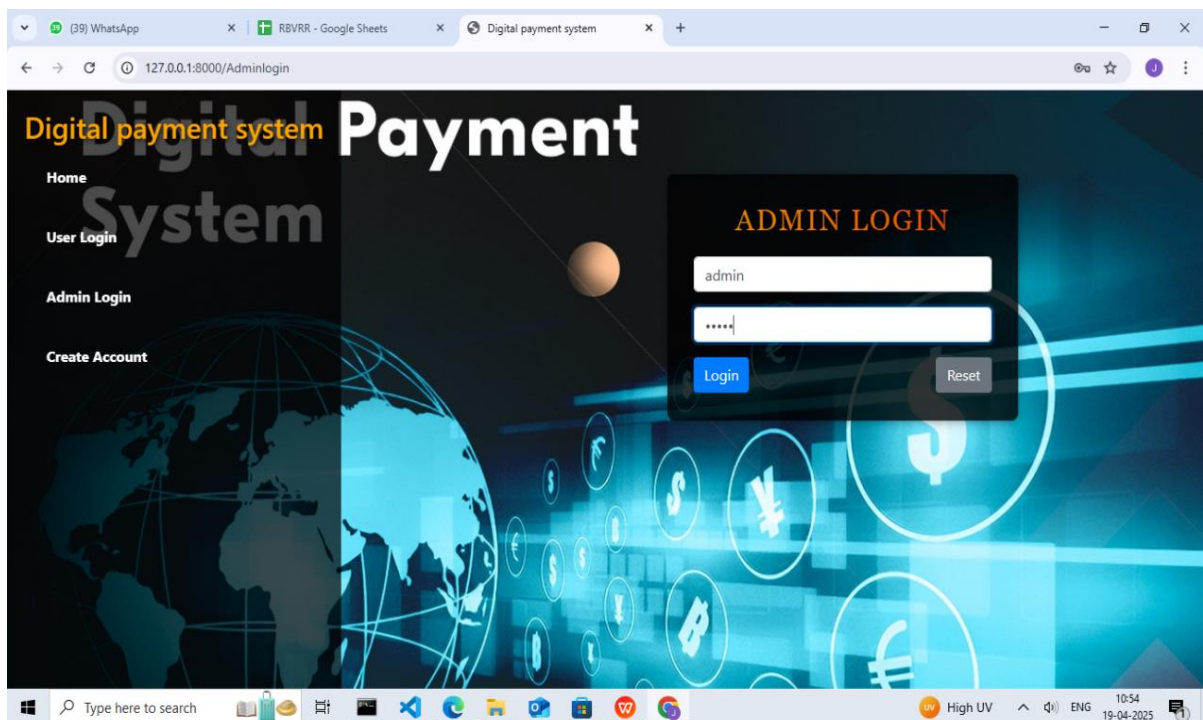
Screen 4.2: User Login Page

This is the page where the Users can login to the portal using their credentials.



Screen 4.3: Admin login Page

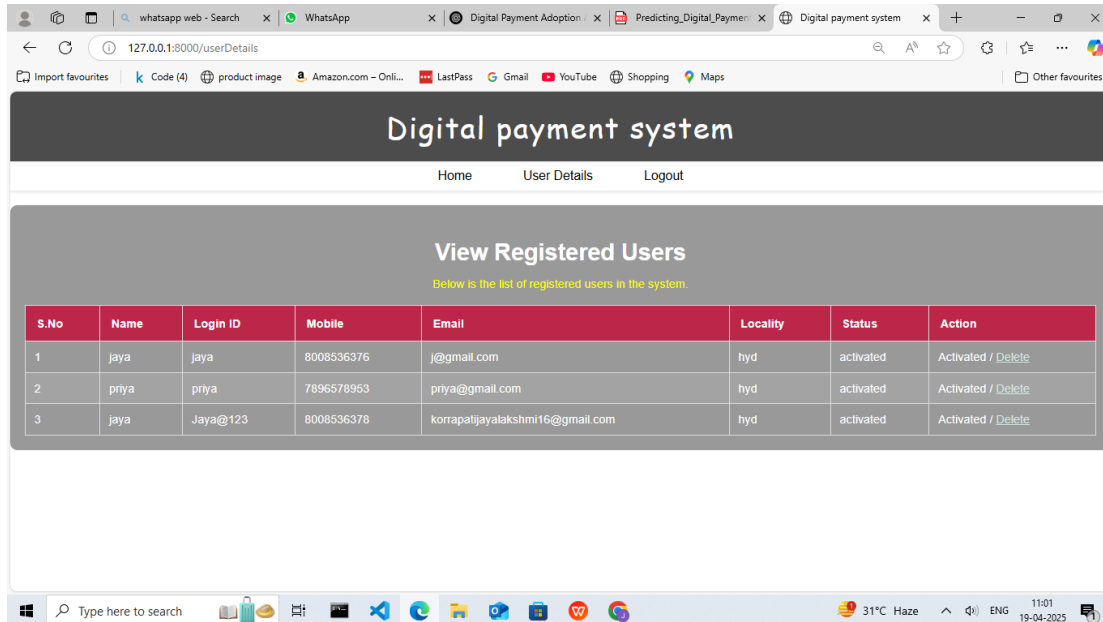
This is the page where the registered user can login to the website using their credentials.



Predicting Digital Payment Platforms Adoption by Retail Vendors Using Machine Learning Techniques

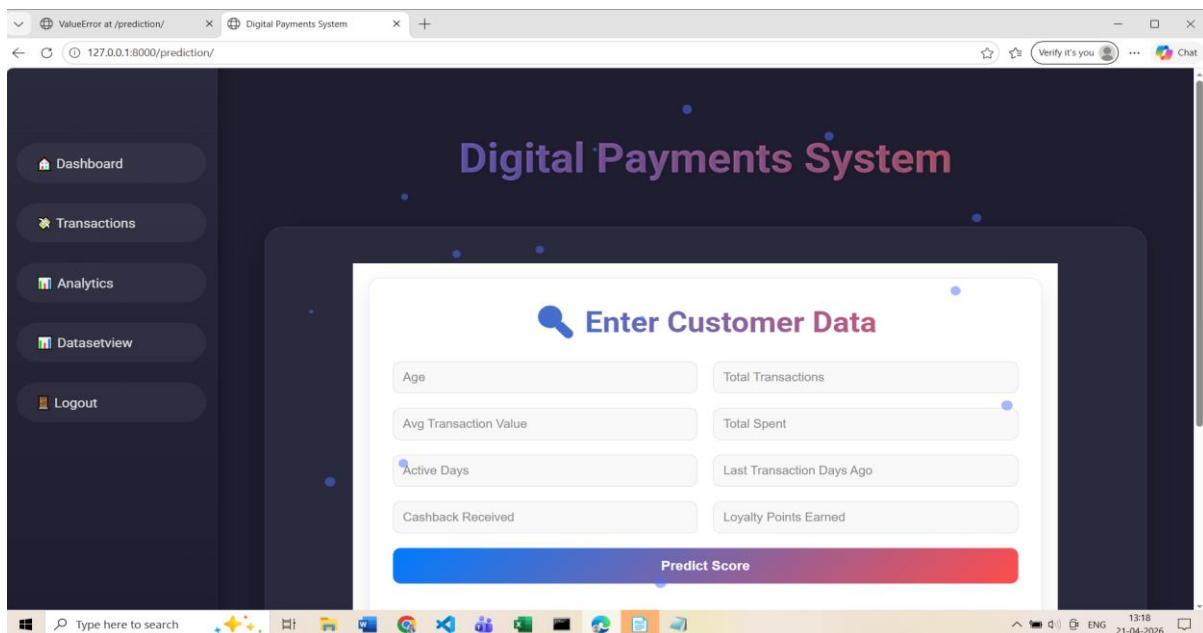
Screen 4.4: User details in admin page

This is the page where the admin can see and manage the user details



Screen 4.5: Vendor Data Input Form

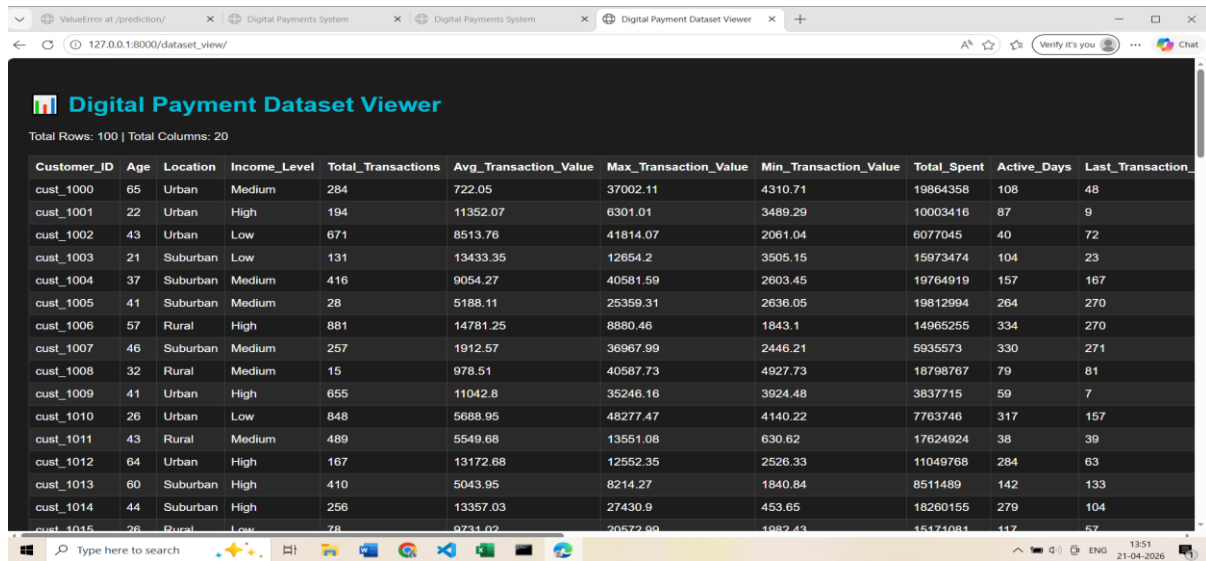
This is the page where the User will give the Vendor Data Inputs



Screen 4.6: Dataset view

This is the page where the Trained data can be seen.

Predicting Digital Payment Platforms Adoption by Retail Vendors Using Machine Learning Techniques

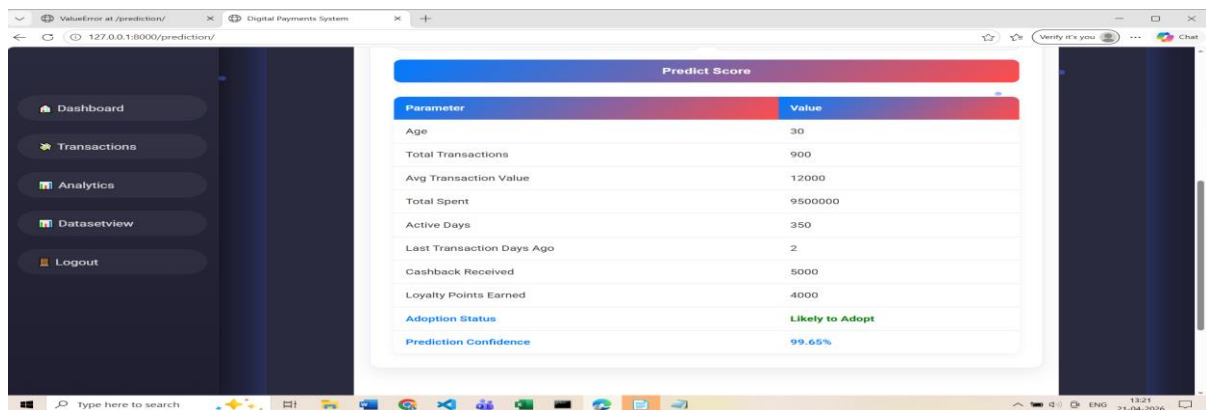


The screenshot displays a web application titled "Digital Payment Dataset Viewer". It shows a table with 100 rows and 20 columns. The columns are: Customer_ID, Age, Location, Income_Level, Total_Transactions, Avg_Transaction_Value, Max_Transaction_Value, Min_Transaction_Value, Total_Spent, Active_Days, and Last_Transaction. The table lists data for various customers, including their age, location (Urban, Suburban, Rural), income level (Low, Medium, High), and transaction metrics.

Customer_ID	Age	Location	Income_Level	Total_Transactions	Avg_Transaction_Value	Max_Transaction_Value	Min_Transaction_Value	Total_Spent	Active_Days	Last_Transaction
cust_1000	65	Urban	Medium	284	722.05	37002.11	4310.71	19864358	108	48
cust_1001	22	Urban	High	194	11352.07	6301.01	3489.29	10003416	87	9
cust_1002	43	Urban	Low	671	8513.76	41814.07	2061.04	6077045	40	72
cust_1003	21	Suburban	Low	131	13433.35	12654.2	3505.15	15973474	104	23
cust_1004	37	Suburban	Medium	416	9054.27	40581.59	2603.45	19764919	157	167
cust_1005	41	Suburban	Medium	28	5188.11	25359.31	2636.05	19812994	284	270
cust_1006	57	Rural	High	881	14781.25	8880.46	1843.1	14965255	334	270
cust_1007	46	Suburban	Medium	257	1912.57	36967.99	2446.21	5935573	330	271
cust_1008	32	Rural	Medium	15	978.51	40587.73	4927.73	18798767	79	81
cust_1009	41	Urban	High	655	11042.8	35246.16	3924.48	3837715	59	7
cust_1010	26	Urban	Low	848	5688.95	48277.47	4140.22	7763746	317	157
cust_1011	43	Rural	Medium	489	5549.68	13551.08	630.62	17624924	38	39
cust_1012	64	Urban	High	167	13172.68	12552.35	2526.33	11049768	284	63
cust_1013	60	Suburban	High	410	5043.95	8214.27	1840.84	8511489	142	133
cust_1014	44	Suburban	High	256	13357.03	27430.9	453.65	18260155	279	104
cust_1015	26	Rural	Low	78	9731.02	20572.89	1082.43	15171081	117	57

Screen 4.7: Prediction

In this stage of the project, the Vendor Input Module serves as the primary data collection interface where business-specific metrics are captured. Users provide critical parameters such as transaction values, cashback history, and loyalty points through a structured dashboard to feed the prediction engine. These real-time inputs are essential for the system to process behavioral patterns and generate an accurate Predicted Satisfaction Score.



The screenshot displays a web application titled "Predict Score". It shows a table with input parameters and their values. The parameters are: Age, Total Transactions, Avg Transaction Value, Total Spent, Active Days, Last Transaction Days Ago, Cashback Received, Loyalty Points Earned, Adoption Status, and Prediction Confidence. The Adoption Status is "Likely to Adopt" and the Prediction Confidence is "99.65%".

Parameter	Value
Age	30
Total Transactions	900
Avg Transaction Value	12000
Total Spent	9500000
Active Days	350
Last Transaction Days Ago	2
Cashback Received	5000
Loyalty Points Earned	4000
Adoption Status	Likely to Adopt
Prediction Confidence	99.65%

4.3.3 Result Analysis

The Support Vector Machine (SVM) and ARIMA models were rigorously trained and tested using vendor behavioral data and historical adoption trends. The evaluation of the system focused on key performance metrics, including Accuracy, Precision, Recall, and the F1-score, to ensure the reliability of the "Likely to Adopt" classifications. The results indicate that the system successfully identifies the core factors—such as perceived usefulness and trust—that

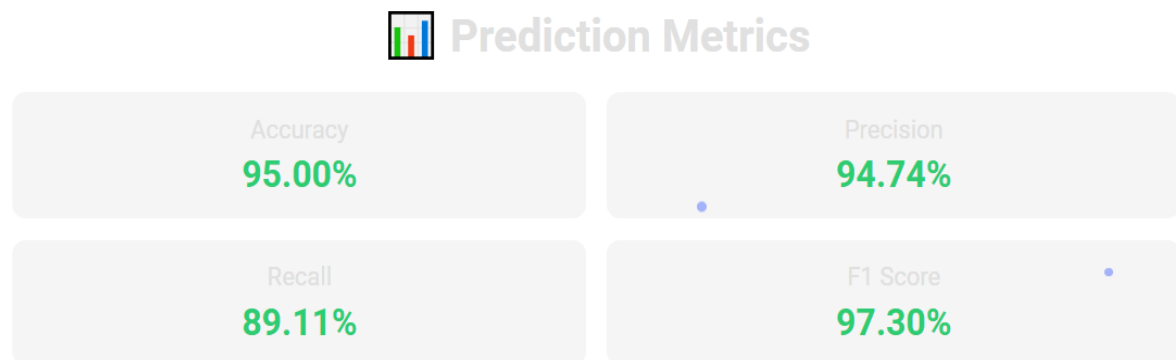


Fig: 4.2 Digital Payment Adoption Evaluation Metrics

5.TESTING AND VALIDATION

5.1 INTRODUCTION

5.1.1 SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

5.1.2 TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centred on the following items:

- Valid Input : identified classes of valid input must be accepted.
- Invalid Input : identified classes of invalid input must be rejected.
- Functions : identified functions must be exercised.
- Output : identified classes of application outputs must be exercised.
- Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

5.2 DESIGN OF EVALUATION METRICS

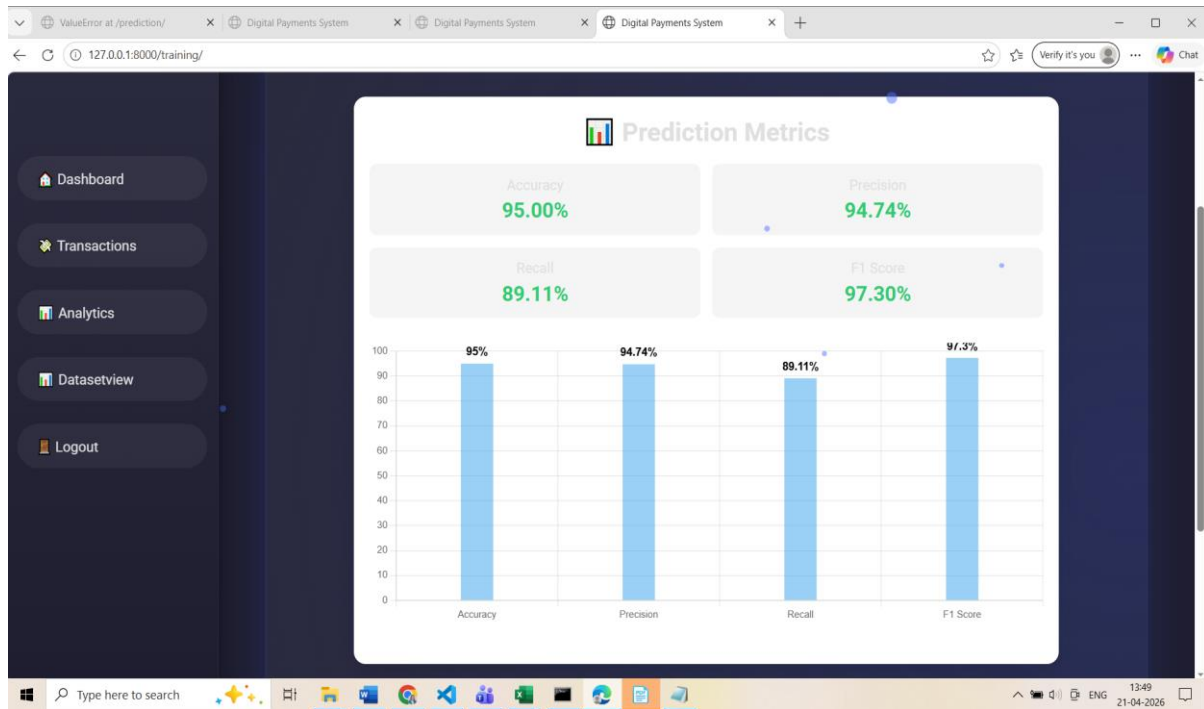


Fig: 5.1 Evaluation Metrics With Bar Graph Visualization

1. Accuracy

Definition:

Accuracy tells how many total predictions were correct out of all predictions.

Formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

2. Precision

Definition:

Precision tells how many predicted positives are actually correct.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

3. Recall (Sensitivity)

Definition:

Recall tells how many actual positives were correctly identified.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

4. F1 Score

Definition:

F1 score is the balance between Precision and Recall (harmonic mean).

Formula:

$$\text{F1 Score} = 2 * [(\text{precision} * \text{Recall}) / (\text{Precision} + \text{Recall})]$$

5.3 VALIDATION

The **Digital Payment Adoption Prediction System** has been thoroughly tested and accurately predicts vendor adoption using SVM and ARIMA models for both real-time and stored data. It handles valid inputs efficiently and provides appropriate error messages for invalid or incomplete data without system failure. Overall, the system is reliable, user-friendly, and ready for real-world deployment.

6.CONCLUSION & FUTURE ENHANCEMENT

6.1 CONCLUSION

This research provides deep insights into what motivates retail vendors in Bangalore to adopt digital payment platforms. By combining traditional statistical analysis with the power of machine learning (specifically SVM), the study identified perceived usefulness, social influence, and compatibility as the leading drivers of adoption. The findings emphasize the necessity for digital platforms to align with existing business practices and peer expectations. Despite moderate accuracy, the SVM model offers promise, and satisfaction analysis shows room for improvement in platforms like GooglePay and AmazonPay. Through forecast modeling with ARIMA, the study shows a steady interest in digital payments, though future variabilities are possible. Moving forward, incorporating more advanced algorithms and qualitative methods can further refine understanding and help stakeholders drive broader digital adoption across the retail sector.

6.2 FUTURE ENHANCEMENT

To elevate the current model's performance and broaden its applicability, several future enhancements are recommended. First, experimenting with advanced machine learning algorithms such as Random Forests, Gradient Boosting, or Neural Networks may uncover more nuanced insights and improve prediction accuracy. Secondly, ensemble methods that combine multiple models can offer improved robustness and reduce bias. Third, incorporating longitudinal data can help capture temporal trends in adoption behavior, offering a more realistic model of vendor behavior over time. Future research should also explore geographical comparisons to identify location-specific motivators or barriers. On the qualitative side, adding interviews, case studies, and focus groups would provide a richer understanding of vendor sentiment and behavior. Lastly, creating a real-time monitoring dashboard using streaming data could help policymakers and digital platform providers to dynamically adjust their strategies and improve vendor onboarding experiences. Use of Advanced ML Algorithms: Introduce models like Random Forest, XGBoost, or Deep Neural Networks for better accuracy and performance.

- Ensemble Modeling: Combine multiple models to improve robustness and reduce overfitting.

- Longitudinal Studies: Track vendor adoption over time to capture temporal changes and behavioral evolution.
- Geographical Comparisons: Expand studies to other cities or states to understand regional variations in adoption.
- Qualitative Research: Add interviews, focus groups, or case studies to extract richer vendor insights beyond numeric data.
- Real-Time Dashboards: Build a live monitoring platform to visualize vendor adoption trends, satisfaction, and prediction metrics in real-time.

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